

Project Design Phase-I

Proposed Solution Template

Date	12 June 2026	Team ID	N/A (Solo Project)
Project Name	Network Anomaly Detection	Maximum Marks	2 Marks

S.No	Parameter	Description
1.	Problem Statement (Problem to be solved)	Traditional network monitoring systems rely on static, rule-based methods that fail to detect unknown or evolving cyber threats such as zero-day attacks, DDoS, and insider threats. Network administrators face an overwhelming volume of traffic data with high false-positive rates, making it difficult to identify genuine anomalies in real-time. There is a critical need for an intelligent, adaptive system that can automatically learn normal network behaviour and flag deviations with minimal human intervention.
2.	Idea / Solution Description	Develop an ML-based Network Anomaly Detection System using the Isolation Forest algorithm trained on the CICIDS 2017 dataset. The system will: • Pre-process and extract key traffic features (packet size, flow duration, protocol type, byte rates, etc.) • Train an Isolation Forest model to learn normal traffic patterns in an unsupervised manner • Flag network flows that deviate significantly from the learned baseline as anomalies • Display results on a simple visualization dashboard showing flagged traffic • Serve as a lightweight, interpretable tool suitable for solo development and academic demonstration
3.	Novelty / Uniqueness	• Unsupervised approach: Uses Isolation Forest which does not require labelled attack data, making it adaptable to new/unknown threats. • Interpretable ML: Unlike black-box deep learning models, Isolation Forest provides explainable anomaly scores that network engineers can understand. • Lightweight & scalable: The model can be retrained on new traffic data without requiring heavy computational resources. • Benchmark comparison: Integrates a statistical baseline (mean + std deviation) alongside the ML model to compare detection accuracy.
4.	Social Impact / Customer Satisfaction	• Enhances cybersecurity posture for organisations of all sizes by proactively detecting threats before major damage occurs. • Reduces financial loss caused by data breaches and network downtime. • Protects personal data of end-users, improving public trust in digital services. • Aids security analysts by reducing alert fatigue through smarter, prioritised anomaly alerts. • Can be extended to protect critical infrastructure such as healthcare, banking, and government networks.
5.	Business Model (Revenue Model)	• SaaS Subscription: Offer the anomaly detection system as a cloud-hosted security service with tiered monthly subscriptions (SMBs, Enterprises). • Freemium Model: Basic anomaly detection free; advanced features (custom dashboards, real-time alerts, API access) under paid plans. • Managed Security Service: Partner with MSSPs (Managed Security Service Providers) to bundle the solution into their offerings. • Consulting & Integration: Charge for custom deployment, integration with existing SIEM tools (Splunk, IBM QRadar), and staff training.

S.No	Parameter	Description
6.	Scalability of the Solution	• The Isolation Forest model can be retrained periodically on new traffic data to adapt to evolving network environments. • Can be containerised using Docker and deployed on cloud platforms (IBM Cloud, AWS, Azure) for horizontal scaling. • The modular architecture allows swapping the detection algorithm (e.g., replacing Isolation Forest with Autoencoder) without redesigning the entire pipeline. • Supports integration with network taps, NetFlow/IPFIX collectors, and SIEM systems to scale from a single network segment to enterprise-wide monitoring. • Dataset can be extended with organisation-specific traffic to improve detection accuracy over time.
